Business Problem

As a marketing agency, our primary objective is to maximize the return on investment (ROI) for our client's advertising campaigns. We have conducted two ad compaigns, one on Facebook and other on AdsWords, and we need to determine which yields better results in terms of clicks, conversion, and overall cost-effectiveness. By idntifying the most effective platform, we can allocate our resources more efficiently and optimize our advertising stratergies to deliver better outcomes for our clients.

Research Question

Which ad platform ismore effective in terms of conversions,, clicks and overall cost-effectiveness?

Importing Libraries

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

import scipy.stats as st
from  sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
from statsmodels.tsa.seasonal import seasonal_decompose
from statsmodels.tsa.stattools import coint

import warnings
warnings.filterwarnings('ignore')
```

Data Description

The dataset comprises a collection of data comparing the performance of two separate ad campaigns conducted throughout the year 2019. Specifically, the data covers a Facebook Ad campaign and an AdWords Ad campaign. For each day of the year 2019, there is a corresponding row in the dataset, resulting in a total of 365 lines of campaign data to analyze. The dataset includes various performance metrics for each ad campaign, providing insights into their effectiveness and efficiency over time.

Key features included in the dataset are as follows:

- Date: The date corresponding to each row of campaign data, ranging from January 1st, 2019, to December 31st, 2019.
- Ad Views: The number of times the ad was viewed.
- Ad Clicks: The number of clicks received on the ad.
- Ad Conversions: The number of conversions resulting from the ad.
- Cost per Ad: The cost associated with running the Facebook ad campaign.
- Click-Through Rate (CTR): The ratio of clicks to views, indicating the effectiveness of the ad in generating clicks.
- Conversion Rate: The ratio of conversions to clicks, reflecting the effectiveness of the ad in driving desired actions.
- Cost per Click (CPC): The average cost incurred per click on the ad.

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
file_path = '/content/drive/MyDrive/Learning/DA-AB Testing/marketing_campaign.csv'
```

```
df = pd.read_csv(file_path)
```

```
df.head()
```

	Date	Facebook Ad Campaign	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions	Cost per Facebook Ad	Facebook Click-Through Rate (Clicks / View)	Facebook Conversion Rate (Conversions / Clicks)	Facebook Cost per Click (Ad Cost / Clicks)	AdWords Ad Campaign	AdWords Ad Views	AdWord A Click
0	1/1/2019	FB_Jan19	2116	18	8	$126	0.83%	42.73%	$7.14	AW_Jan19	4984	5
1	1/2/2019	FB_Jan19	3106	36	12	$104	1.15%	34.04%	$2.91	AW_Jan19	4022	7
2	1/3/2019	FB_Jan19	3105	26	8	$102	0.84%	31.45%	$3.89	AW_Jan19	3863	4
3	1/4/2019	FB_Jan19	1107	27	9	$71	2.45%	34.76%	$2.62	AW_Jan19	3911	4

Next steps: Generate code with df New interactive sheet

```
# rows and columns count of the dataset
df.shape
```

```
(365, 17)
```

```
# datatypes of the columns
df.dtypes
```

	0
Date	object
Facebook Ad Campaign	object
Facebook Ad Views	int64
Facebook Ad Clicks	int64
Facebook Ad Conversions	int64
Cost per Facebook Ad	object
Facebook Click-Through Rate (Clicks / View)	object
Facebook Conversion Rate (Conversions / Clicks)	object
Facebook Cost per Click (Ad Cost / Clicks)	object
AdWords Ad Campaign	object
AdWords Ad Views	int64
AdWords Ad Clicks	int64
AdWords Ad Conversions	int64
Cost per AdWords Ad	object
AdWords Click-Through Rate (Clicks / View)	object
AdWords Conversion Rate (Conversions / Click)	object
AdWords Cost per Click (Ad Cost / Clicks)	object

dtype: object

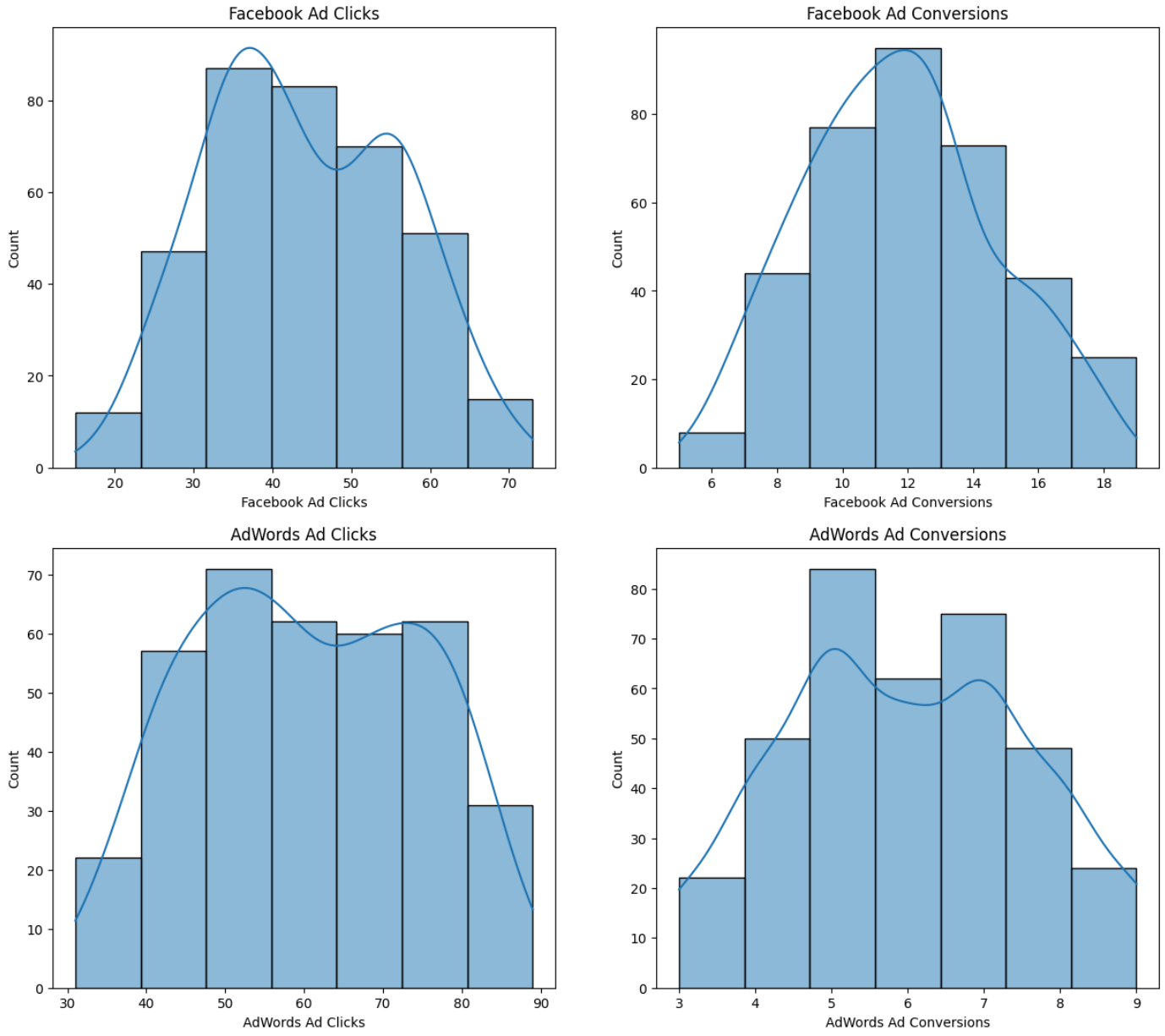
```
# converting date to datetime
df['Date'] = pd.to_datetime(df['Date'])
```

```
# descriptive stats of the campaigns
df.describe()
```

	Date	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions	AdWords Ad Views	AdWords Ad Clicks	AdWords Ad Conversions
count	365	365.000000	365.000000	365.000000	365.00000	365.000000	365.000000
mean	2019-07-02 00:00:00	2179.687671	44.049315	11.742466	4717.19726	60.383562	5.980822
min	2019-01-01 00:00:00	1050.000000	15.000000	5.000000	3714.00000	31.000000	3.000000
25%	2019-04-02 00:00:00	1656.000000	35.000000	10.000000	4247.00000	49.000000	5.000000
50%	2019-07-02 00:00:00	2202.000000	43.000000	12.000000	4711.00000	60.000000	6.000000

Comparing Campaigns performance

```
# distribution of the clicks and conversions
plt.figure(figsize=(15,6))
plt.subplot(1,2,1)
plt.title('Facebook Ad Clicks')
sns.histplot(df['Facebook Ad Clicks'], bins = 7, edgecolor = 'k', kde = True)
plt.subplot(1,2,2)
plt.title('Facebook Ad Conversions')
sns.histplot(df['Facebook Ad Conversions'], bins = 7, edgecolor = 'k', kde = True)
plt.show()


plt.figure(figsize=(15,6))
plt.subplot(1,2,1)
plt.title('AdWords Ad Clicks')
sns.histplot(df['AdWords Ad Clicks'], bins = 7, edgecolor = 'k', kde = True)
plt.subplot(1,2,2)
plt.title('AdWords Ad Conversions')
sns.histplot(df['AdWords Ad Conversions'], bins = 7, edgecolor = 'k', kde = True)
plt.show()
```



All the histogram are showing somewhat symmetrical shape. This symmetrical shape suggests that the number of clicks and conversions is relatively evenly distributed. In other words, there are not many clicks or conversions that are outliers on either the high or low end.

How frequently do we observe days with high numbers of conversions compared to days with low numbers of conversions?

```
# creating function to calculate the category for the conversions
def create_conversion_category(conversion_col):
    category = []
    for conversion in df[conversion_col]:
        if conversion < 6:
            category.append('less than 6')
        elif 6 <= conversion < 11:
            category.append('6 - 10')
        elif 11 <= conversion < 16:
            category.append('10 - 15')
        else:
```

```
            category.append('more than 15')
    return category

# applying function of different campaign's conversions
df['Facebook Conversion Category'] = create_conversion_category('Facebook Ad Conversions')
df['AdWords Conversion Category'] = create_conversion_category('AdWords Ad Conversions')
```

```
df[['Facebook Ad Conversions','Facebook Conversion Category','AdWords Ad Conversions','AdWords Conversion Category']].head(
```

	Facebook Ad Conversions	Facebook Conversion Category	AdWords Ad Conversions	AdWords Conversion Category
0	8	6 - 10	5	less than 6
1	12	10 - 15	6	6 - 10
2	8	6 - 10	4	less than 6
3	9	6 - 10	5	less than 6
4	7	6 - 10	7	6 - 10

```
df['Facebook Conversion Category'].value_counts()
```

Facebook Conversion Category	count
10 - 15	189
6 - 10	128
more than 15	47
less than 6	1

dtype: int64

```
facebook = pd.DataFrame(df['Facebook Conversion Category'].value_counts()).reset_index().rename(columns = {'Facebook Conver
facebook
```

	Category	count
0	10 - 15	189
1	6 - 10	128
2	more than 15	47
3	less than 6	1

Next steps: Generate code with facebook New interactive sheet

```
df['AdWords Ad Conversions'].value_counts()
```

AdWords Ad Conversions	count
5	84
7	75
6	62
4	50
8	48
9	24
3	22

dtype: int64

```
df['AdWords Conversion Category'].value_counts()
```

AdWords Conversion Category	count
6 - 10	209
less than 6	156

dtype: int64

```
adwords = pd.DataFrame(df['AdWords Conversion Category'].value_counts()).reset_index().rename(columns = {'AdWords Conversio
adwords
```

	Category	count
0	6 - 10	209
1	less than 6	156

Next steps: Generate code with adwords New interactive sheet

```
category_df = pd.merge(facebook, adwords, on = 'Category', how = 'outer').fillna(0)
category_df
```

	Category	count_x	count_y
0	10 - 15	189	0.0
1	6 - 10	128	209.0
2	less than 6	1	156.0
3	more than 15	47	0.0

Next steps: Generate code with category_df New interactive sheet

```
category_df = category_df.iloc[[3,1,0,2]]
category_df
```

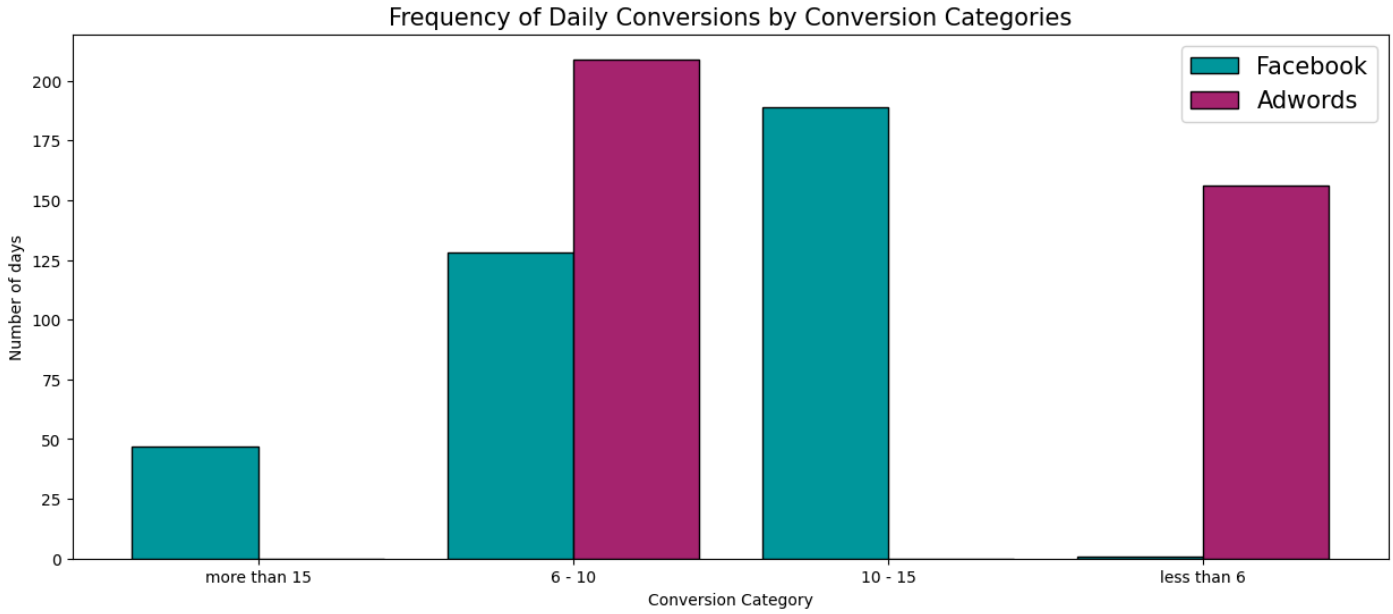
	Category	count_x	count_y
3	more than 15	47	0.0
1	6 - 10	128	209.0
0	10 - 15	189	0.0
2	less than 6	1	156.0

Next steps: Generate code with category_df New interactive sheet

```
X_axis = np.arange(len(category_df))
X_axis
```

```
array([0, 1, 2, 3])
```

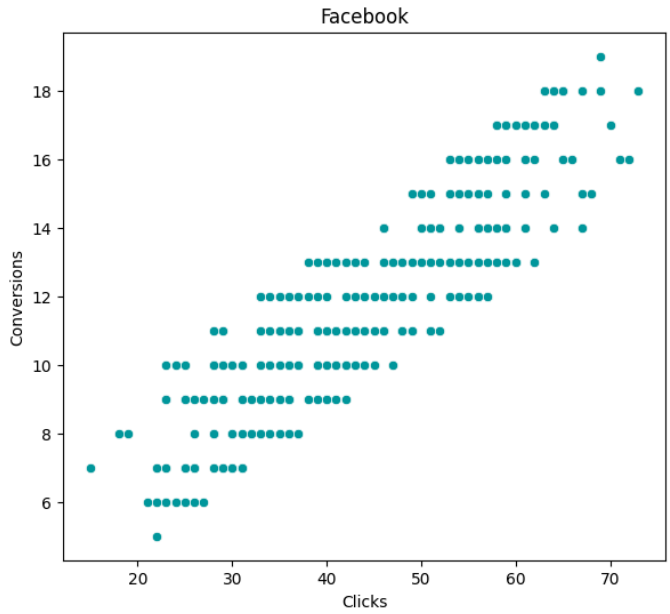
```
X_axis = np.arange(len(category_df))
plt.figure(figsize = (15,6))
plt.bar(X_axis - 0.2, category_df['count_x'], 0.4, label = 'Facebook', color = '#03989E', linewidth = 1, edgecolor = 'k')
plt.bar(X_axis + 0.2, category_df['count_y'], 0.4, label = 'Adwords', color = '#A62372', linewidth = 1, edgecolor = 'k')

plt.xticks(X_axis, category_df['Category'])
plt.xlabel("Conversion Category")
plt.ylabel("Number of days")
plt.title("Frequency of Daily Conversions by Conversion Categories", fontsize = 15)
plt.legend(fontsize = 15)
plt.show()
```

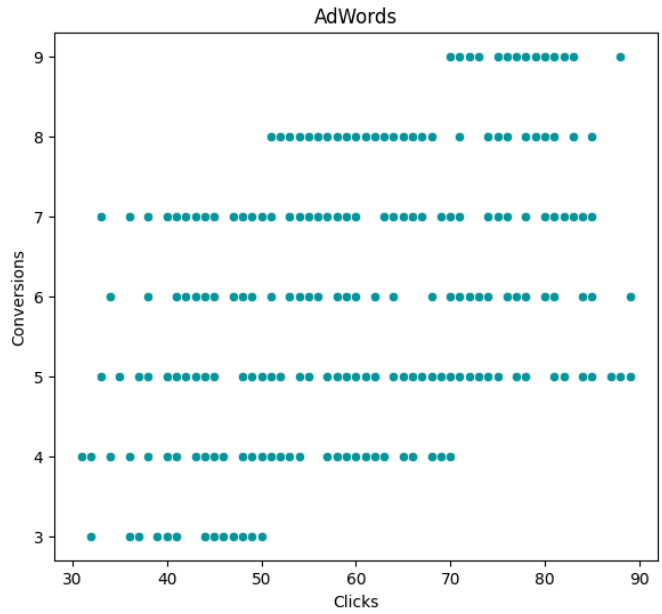


- The data suggests Facebook had more frequent higher conversion days than AdWords, which either had very low conversion rates (less than 6) or moderate ones (6 - 10).
- There is a significant variance in the number of high-conversion days between two different campaigns.
- The absence of any days with conversions between 10 - 15 and more than 15 in AdWords indicates a need to review what strategies were changed or what external factors could have influenced these numbers.

Do more clicks on the ad really lead to more sales?

```
plt.figure(figsize=(15,6))
plt.subplot(1,2,1)
plt.title('Facebook')
sns.scatterplot(x = df['Facebook Ad Clicks'],y = df['Facebook Ad Conversions'], color = '#03989E')
plt.xlabel('Clicks')
plt.ylabel('Conversions')
plt.subplot(1,2,2)
plt.title('AdWords')
sns.scatterplot(x = df['AdWords Ad Clicks'],y = df['AdWords Ad Conversions'], color = '#03989E')
plt.xlabel('Clicks')
plt.ylabel('Conversions')
plt.show()
```






```
facebook_corr = df[['Facebook Ad Conversions','Facebook Ad Clicks']].corr()
facebook_corr
```

	Facebook Ad Conversions	Facebook Ad Clicks
Facebook Ad Conversions	1.000000	0.873775
Facebook Ad Clicks	0.873775	1.000000

Next steps: Generate code with facebook_corr New interactive sheet

```
adwords_corr = df[['AdWords Ad Conversions','AdWords Ad Clicks']].corr()
adwords_corr
```

	AdWords Ad Conversions	AdWords Ad Clicks
AdWords Ad Conversions	1.000000	0.447993
AdWords Ad Clicks	0.447993	1.000000

Next steps: Generate code with adwords_corr New interactive sheet

```
print('Correlation Coeff \n--------------')
print('Facebook :',round(facebook_corr.values[0,1],2))
print('AdWords : ',round(adwords_corr.values[0,1],2))
```

```
Correlation Coeff
--------------
Facebook : 0.87
AdWords :  0.45
```

- A correlation coefficient of 0.87 indicates a strong positive linear relationship between clicks on Facebook ads and sales. This suggests that as the number of clicks on Facebook ads increases, sales tend to increase as well.
- This strong correlation suggests that Facebook ads are highly effective in driving sales, as a large portion of the variation in sales can be explained by the variation in clicks on Facebook ads.
- The strong correlation between clicks on Facebook ads and sales suggests that Facebook advertising is highly effective in driving sales for the business. Increasing investment in Facebook ads or optimizing their performance could potentially lead to even higher sales.
- A correlation coefficient of 0.45 indicates a moderate positive linear relationship between clicks on AdWords ads and sales. While there is still a positive relationship, it is not as strong as with Facebook ads.
- The moderate correlation between clicks on AdWords ads and sales indicates that while AdWords advertising does contribute to sales, its effectiveness may be influenced by other factors. Further analysis is needed to identify these factors and optimize AdWords campaigns accordingly.

Hypothesis Testing

Hypothesis: Advertising on Facebook will result in a greater number of conversions compared to advertising on AdWords.

Null Hypothesis (H0): There is no difference in the number of conversions between Facebook and AdWords, or the number of conversions from AdWords is greater than or equal to those from Facebook.

H0: μ_Facebook ≤ μ_AdWords

Alternate Hypothesis (H1): The number of conversions from Facebook is greater than the number of conversions from AdWords.

H1: μ_Facebook > μ_AdWords

```
print('Mean Conversion \n--------------')
print('Facebook :', round(df['Facebook Ad Conversions'].mean(),2))
print('AdWords :', round(df['AdWords Ad Conversions'].mean(),2))

t_stats, p_value = st.ttest_ind(a = df['Facebook Ad Conversions'], b = df['AdWords Ad Conversions'], equal_var = False)
print('\nT statistic', t_stats, '\np-value',p_value)

# comparing the p value with the significance of 5% or 0.05
if p_value < 0.05:
    print("\np-value is less than significance value, Reject the null hypothesis")
else:
    print("\np-value is greater than significance value, Accept the null hypothesis")
```

```
Mean Conversion
--------------
Facebook : 11.74
AdWords : 5.98

T statistic 32.88402060758184
p-value 9.348918164530465e-134

p-value is less than significance value, Reject the null hypothesis
```

- The mean number of conversions from Facebook ads (11.74) is substantially higher than the mean number of conversions from AdWords ads (5.98). This suggests that, on average, Facebook advertising is more effective in generating conversions compared to AdWords advertising.
- The T statistic (32.88) is a measure of the difference between the means of the two groups relative to the variation within the groups. A larger T statistic indicates a greater difference between the means of the two groups.
- The p-value (9.35e-134) is extremely small, indicating strong evidence against the null hypothesis.
- The results strongly support the alternate hypothesis, indicating that the number of conversions from Facebook advertising is indeed greater than the number of conversions from AdWords advertising.
- Facebook advertising appears to be a more effective channel for generating conversions compared to AdWords advertising, based on the sample data analyzed.
- Given the significant difference in conversion rates between Facebook and AdWords, consider reallocating resources towards Facebook advertising efforts. This could involve increasing ad spend, expanding targeting efforts, or experimenting with different ad formats to capitalize on the platform's effectiveness in driving conversions.

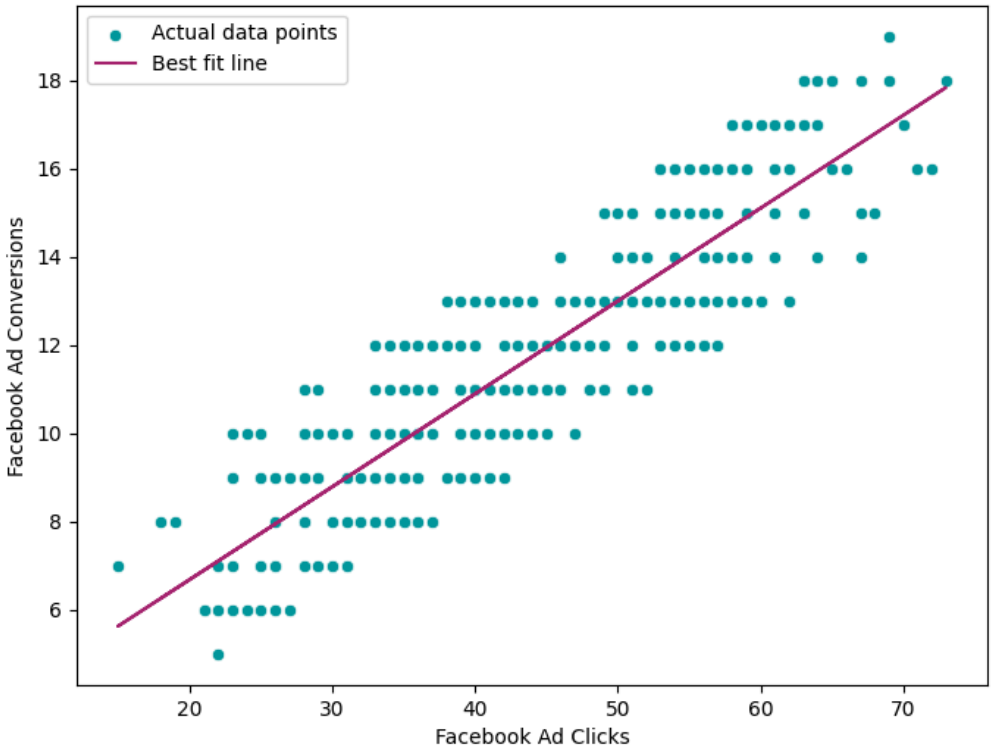
Regression Analysis

What will happen when I do go with the Facebook Ad? How many facebook ad conversions can I expect given a certain number of facebook ad clicks?

```
# independent variable
X = df[['Facebook Ad Clicks']]

# dependent variable
y = df[['Facebook Ad Conversions']]

# initializing and fitting Linear Regression model
reg_model = LinearRegression()
reg_model.fit(X,y)
prediction = reg_model.predict(X)

# model evaluation
r2 = r2_score(y, prediction)*100
mse = mean_squared_error(y, prediction)
print('Accuracy (R2 Score):',round(r2,2),'%')
print('Mean Squared Error:', round(mse,2))
```

```
Accuracy (R2 Score): 76.35 %
Mean Squared Error: 2.02
```

```
plt.figure(figsize=(8,6))
sns.scatterplot(x = df['Facebook Ad Clicks'],y = df['Facebook Ad Conversions'], color = '#03989E', label = 'Actual data point
plt.plot(df['Facebook Ad Clicks'], prediction, color = '#A62372', label = 'Best fit line')
plt.legend()
plt.show()
```




```
print(f'For {50} Clicks, Expected Conversion : {round(reg_model.predict([[50]])[0][0],2)}')
print(f'For {80} Clicks, Expected Conversion : {round(reg_model.predict([[80]])[0][0],2)}')
```

```
For 50 Clicks, Expected Conversion : 13.0
For 80 Clicks, Expected Conversion : 19.31
```

- The model has a reasonably good predictive power, with an R2 score of 76.35%. This suggests that it can effectively predict Facebook ad conversions based on the number of Facebook ad clicks.
- With the insights provided by the Linear Regression model, businesses can make informed decisions about resource allocation, budget planning, and campaign optimization.
- For instance, knowing the expected number of Facebook ad conversions based on a certain number of Facebook ad clicks can help in setting realistic campaign goals, optimizing ad spend, and assessing the ROI of Facebook advertising efforts.

Analyzing Facebook Campaign metrics over time.

```
# cleaning data ( removing unwanted symbols from the columns and converting them to numerical columns)
df['Facebook Click-Through Rate (Clicks / View)'] = df['Facebook Click-Through Rate (Clicks / View)'].apply(lambda x: float
df['Facebook Conversion Rate (Conversions / Clicks)'] = df['Facebook Conversion Rate (Conversions / Clicks)'].apply(lambda
df['Facebook Cost per Click (Ad Cost / Clicks)'] = df['Facebook Cost per Click (Ad Cost / Clicks)'].apply(lambda x: float(x
df['Cost per Facebook Ad'] = df['Cost per Facebook Ad'].apply(lambda x: float(x[1:]))
```

```
# filtering for facebook campaign
df = df[['Date','Facebook Ad Views',
       'Facebook Ad Clicks', 'Facebook Ad Conversions', 'Cost per Facebook Ad',
       'Facebook Click-Through Rate (Clicks / View)',
       'Facebook Conversion Rate (Conversions / Clicks)',
       'Facebook Cost per Click (Ad Cost / Clicks)']]

df.head()
```

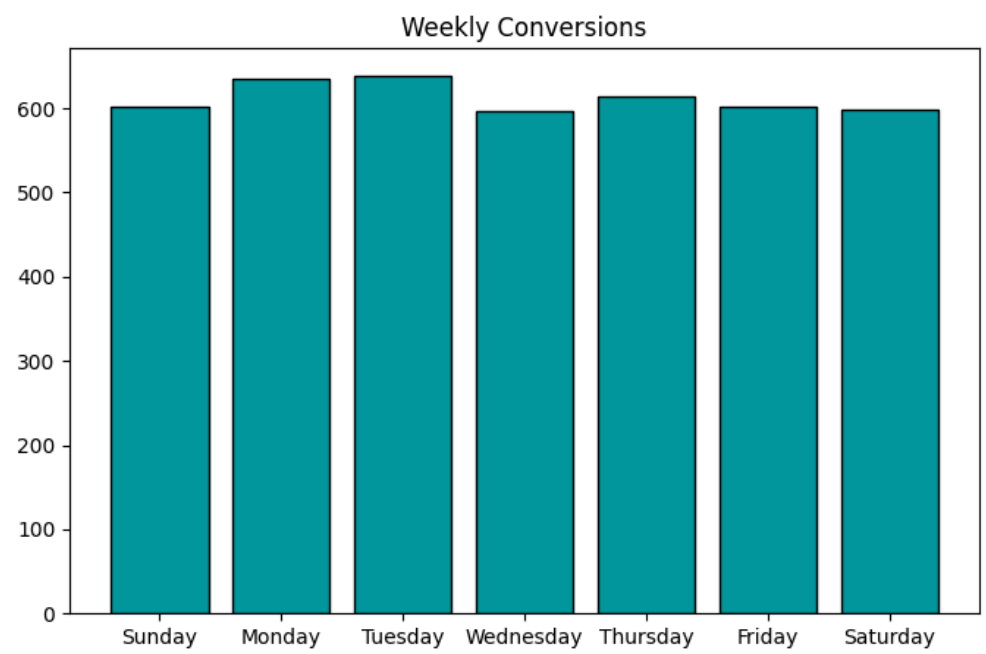
	Date	Facebook Ad Views	Facebook Ad Clicks	Facebook Ad Conversions	Cost per Facebook Ad	Facebook Click-Through Rate (Clicks / View)	Facebook Conversion Rate (Conversions / Clicks)	Facebook Cost per Click (Ad Cost / Clicks)
0	2019-01-01	2116	18	8	126.0	0.83	42.73	7.14
1	2019-01-02	3106	36	12	104.0	1.15	34.04	2.91
2	2019-01-03	3105	26	8	102.0	0.84	31.45	3.89
3	2019-01-04	1107	27	9	71.0	2.45	34.76	2.62
4	2019-01-05	1317	15	7	78.0	1.10	47.59	5.38

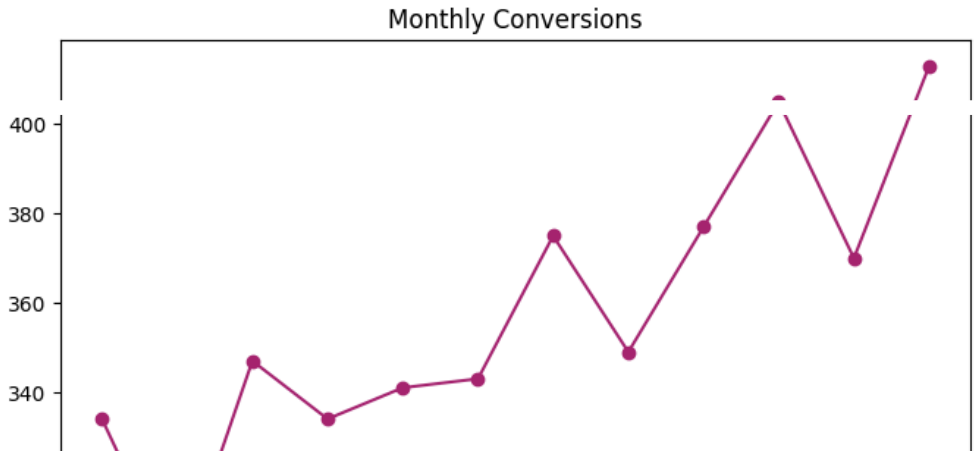
Next steps: Generate code with df New interactive sheet

At what times of the month or days of the week do we observe the conversions?

```
# extracting month and week day from the date column
df['month'] = df['Date'].dt.month
df['week'] = df['Date'].dt.weekday
```

```
plt.figure(figsize=(8,5))
plt.title('Weekly Conversions')
weekly_conversion = df.groupby('week')[['Facebook Ad Conversions']].sum()
week_names= ['Sunday','Monday','Tuesday','Wednesday','Thursday','Friday','Saturday']
plt.bar(week_names, weekly_conversion['Facebook Ad Conversions'], color = '#03989E', edgecolor = 'k')
plt.show()
```




```
plt.figure(figsize=(8,5))
plt.title('Monthly Conversions')
monthly_conversion = df.groupby('month')[['Facebook Ad Conversions']].sum()
month_names = ['Jan','Feb','Mar','Apr','May','Jun','Jul','Aug','Sep','Oct','Nov','Dec']
```

```
plt.plot(month_names, monthly_conversion['Facebook Ad Conversions'],'-o', color = '#A62372')
```